Food and Agriculture Organization of the United Nations

Climate change and climate change impacts on São Tomé and Príncipe

with a focus on fishery and tourism sectors in a Blue Economy context

Sao Tome and Principe, November/December 2025

GCF PROJECT CYCLE AND REPORT FINDINGS

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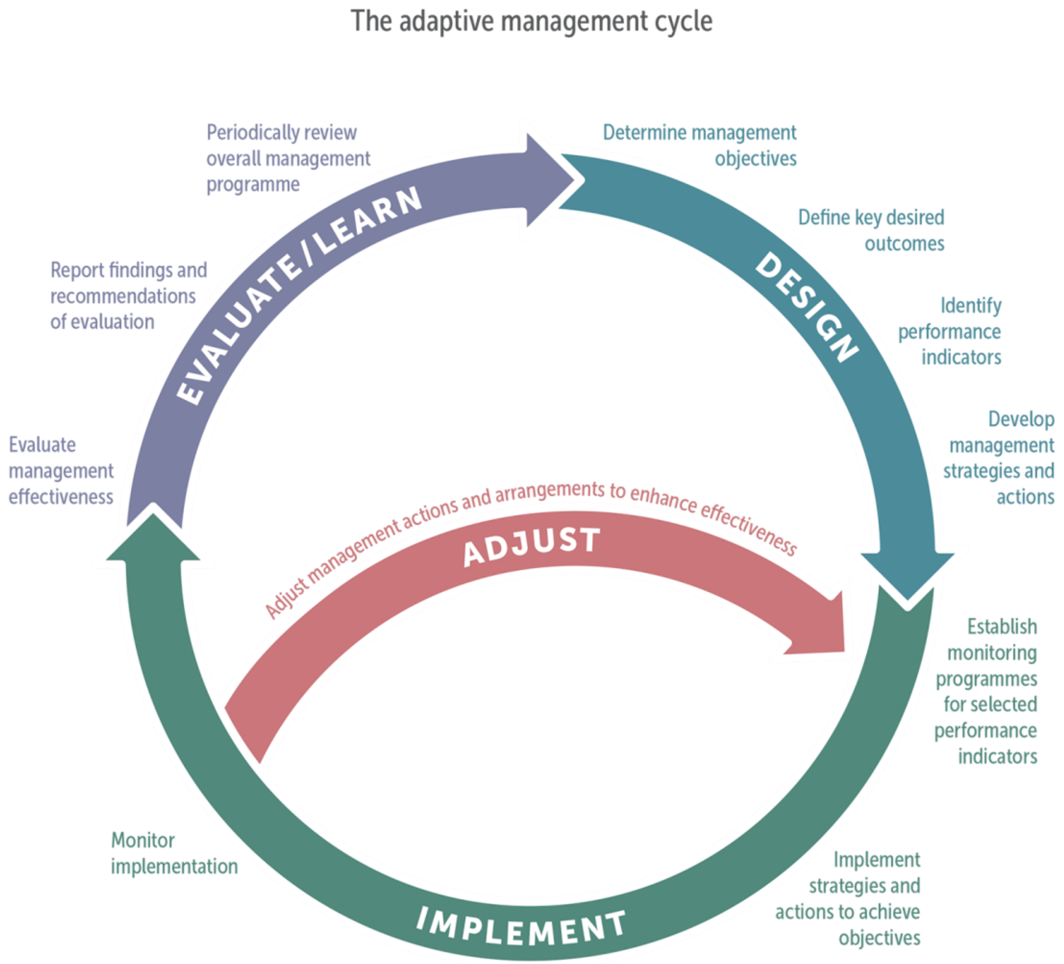
The GCF process



Source: Adapted from DPIPWE (2014) after Jones (2005, 2009).

Report overview

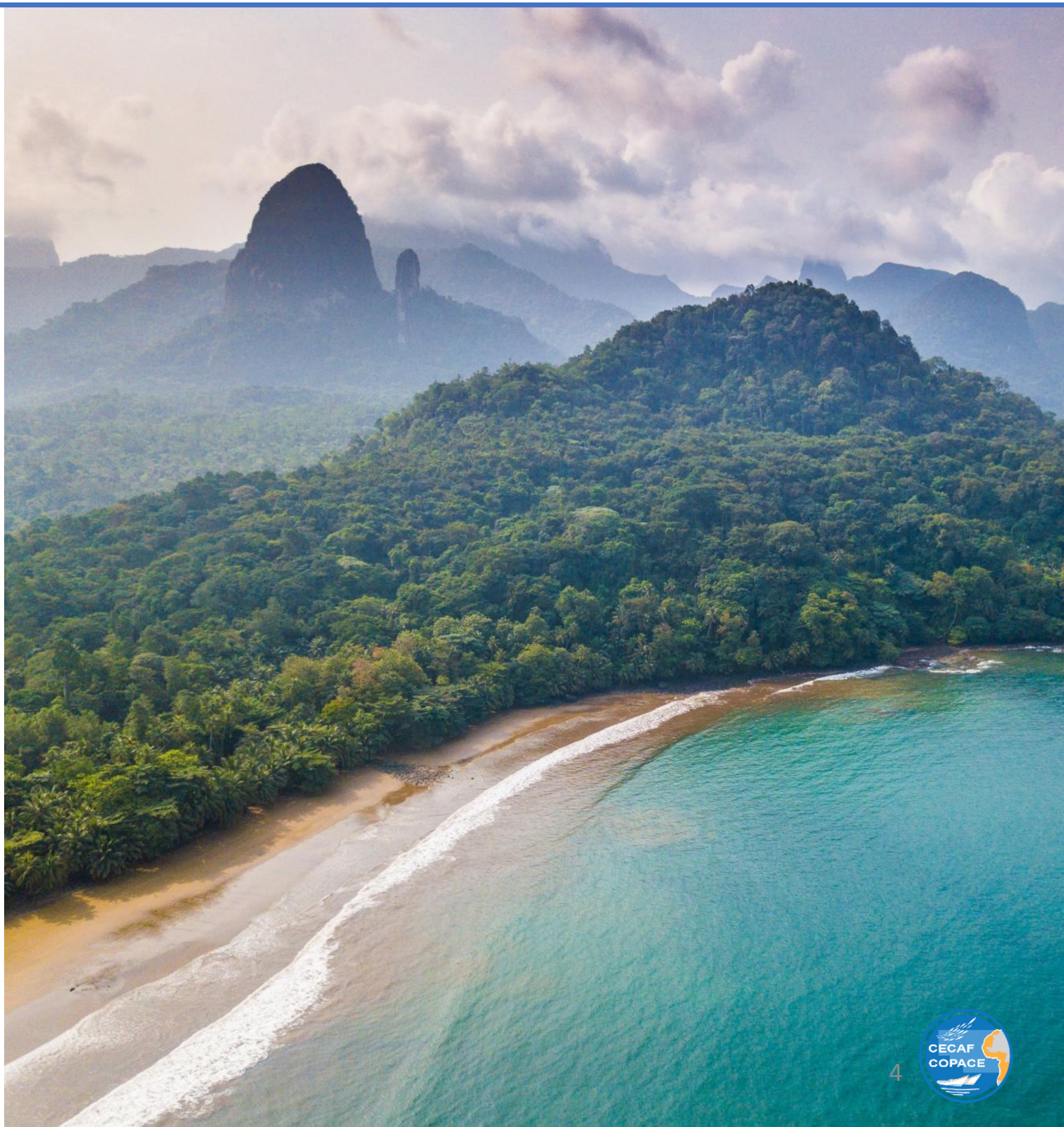
- Assess climate impact on STP fisheries and tourism:

- Climate exposure
- Climate vulnerability
- Climate adaptation

Tourism in STP

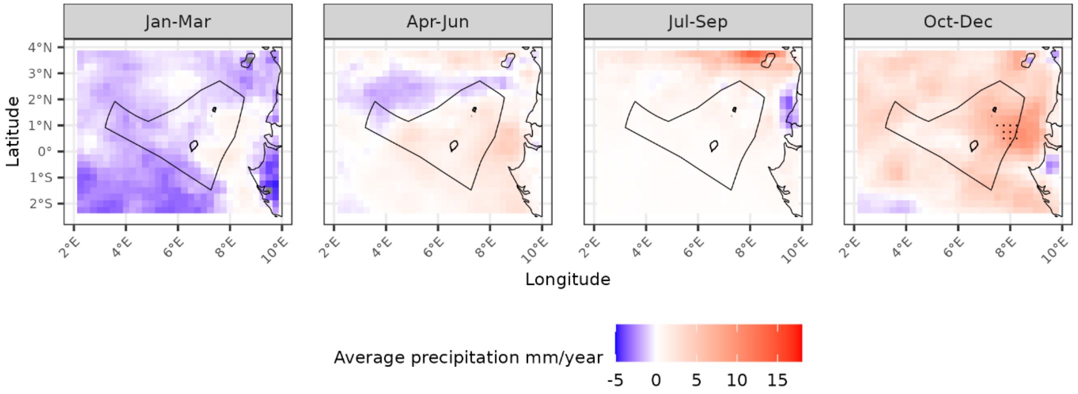
- GDP: 6-8% (14% indirect)
- Employment: 10%
- Highlights:
  - Sun and beach
  - Diving
  - Ecotourism
  - *Sports fishing*

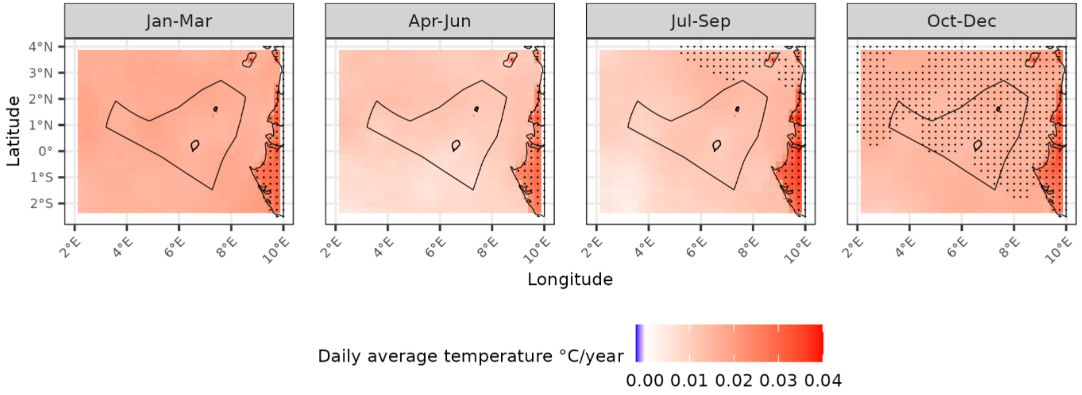
- Threats
  - Sea level rise
  - Tourist discomfort

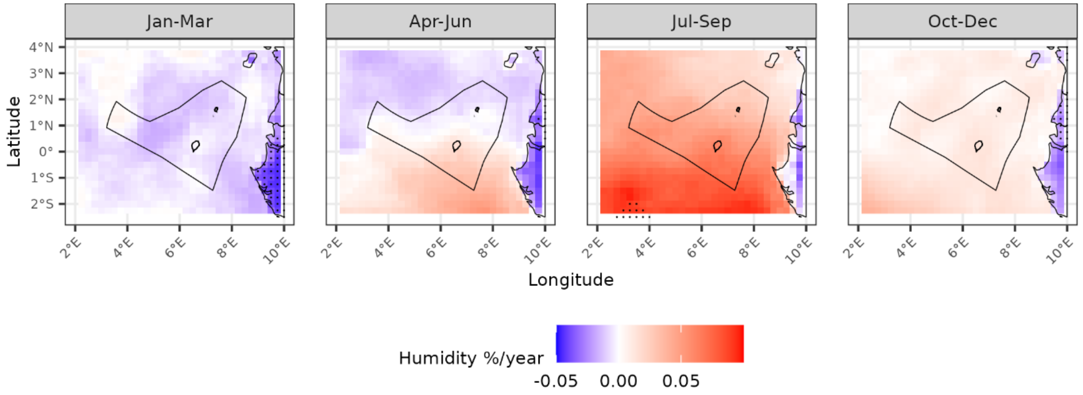


Tourism – climate exposure

- Past climate trends:
  - *Hotter – increasing temperatures and heatwave frequency*
  - *Wetter – increasing precipitation in Oct-Dec and more high precipitation events*
  - *More humid in Jul-Dec*



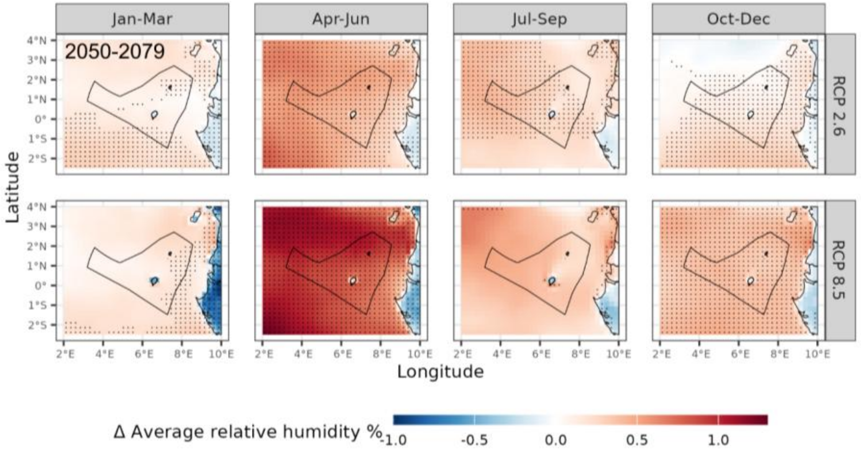


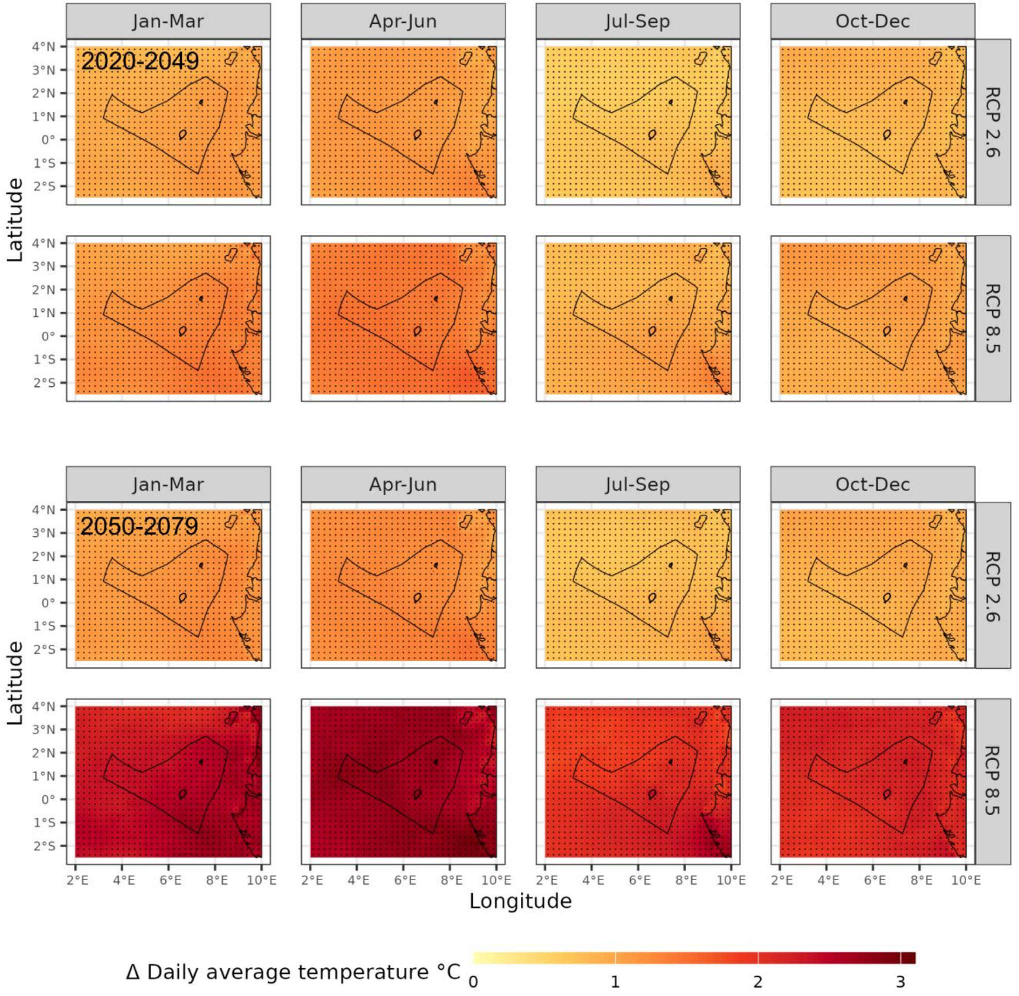


Tourism – climate exposure

- Future climate trends:
  - *Hotter – especially in Jan-June*
    - *Higher warming by 2050-2079*
    - *Higher warming under RCP 8.5*

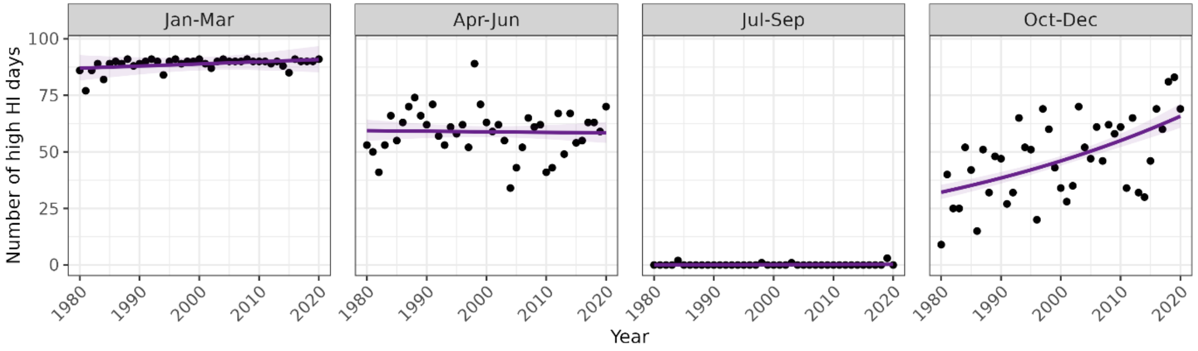
  - *More precipitation in Apr-Dec*
  - *More humid in Apr-Jun*





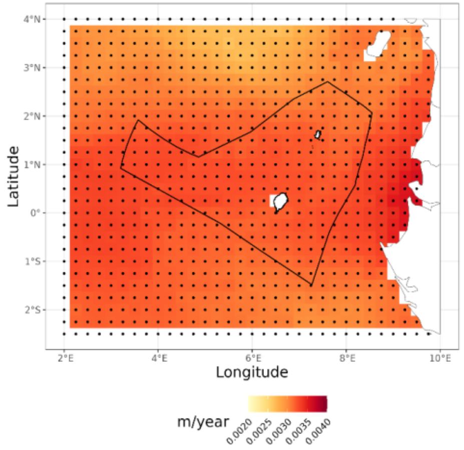
Tourism – climate impact

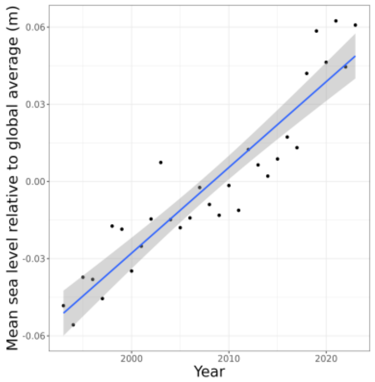
- Historical increase in high heat index days in Oct-Dec



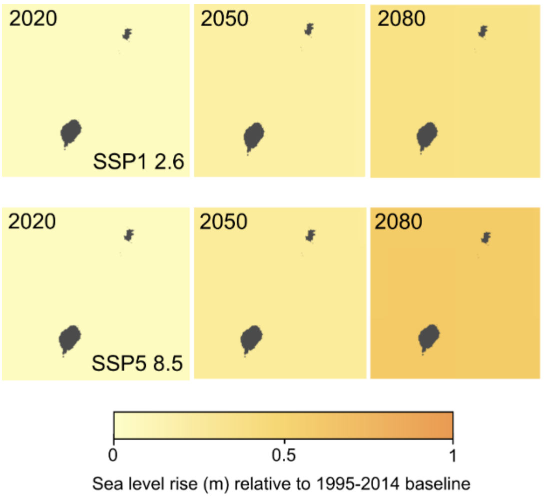
- *Historical and projected sea level rise*

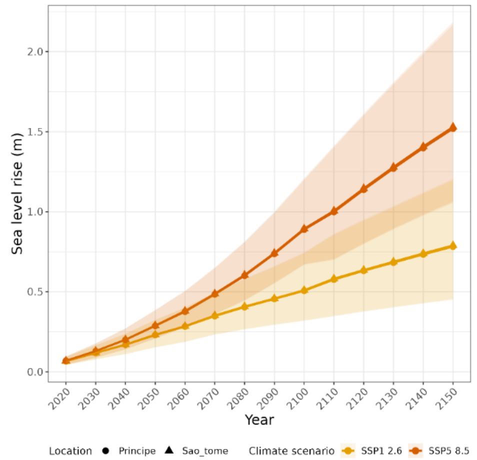
Historical





Projected





Fisheries in STP

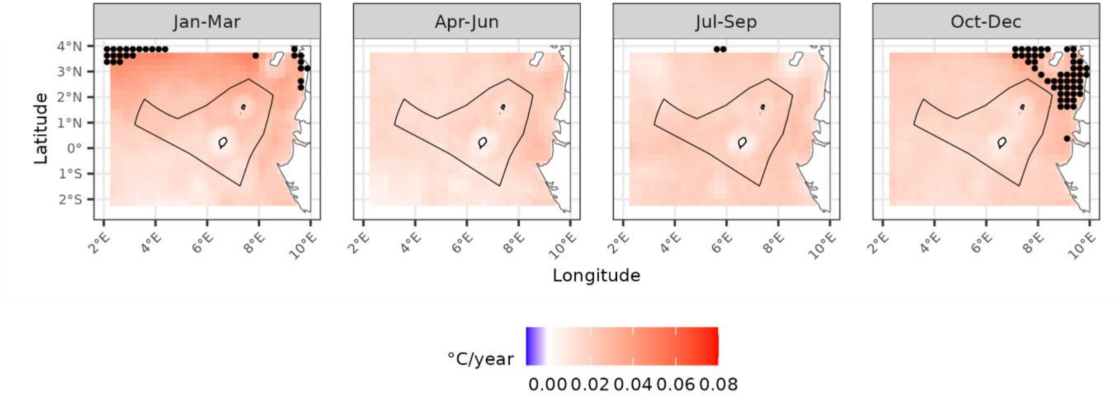
- Fisheries are crucial for the economy and employment
  - 5-10% GDP
  - Employ 10-15% population
  - Fish comprises ~53% of animal protein intake

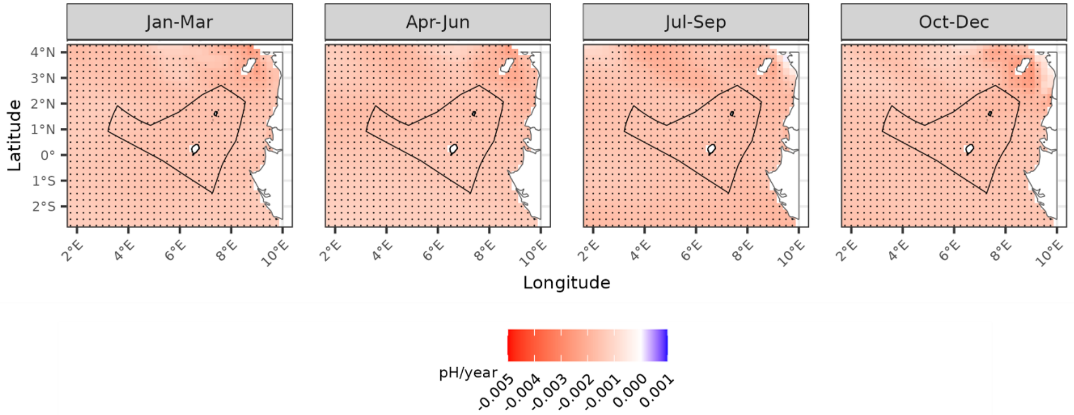
- Fish conservation is also important for future tourism:
  - Sports fishing
  - Diving

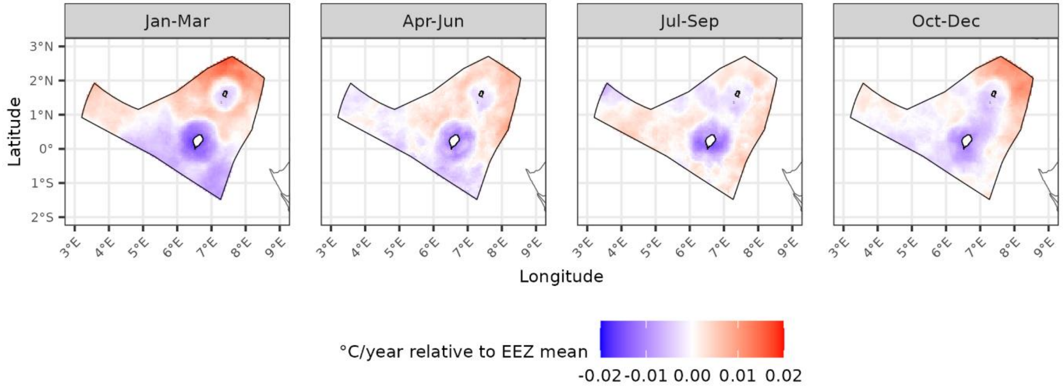
- Threats
  - Ocean warming
  - Ocean acidification

Fisheries – climate exposure

- Past climate trends:
  - *Hotter – increasing temperatures and marine heatwave frequency*
  - *Acidification*
  - *Deoxygenation*

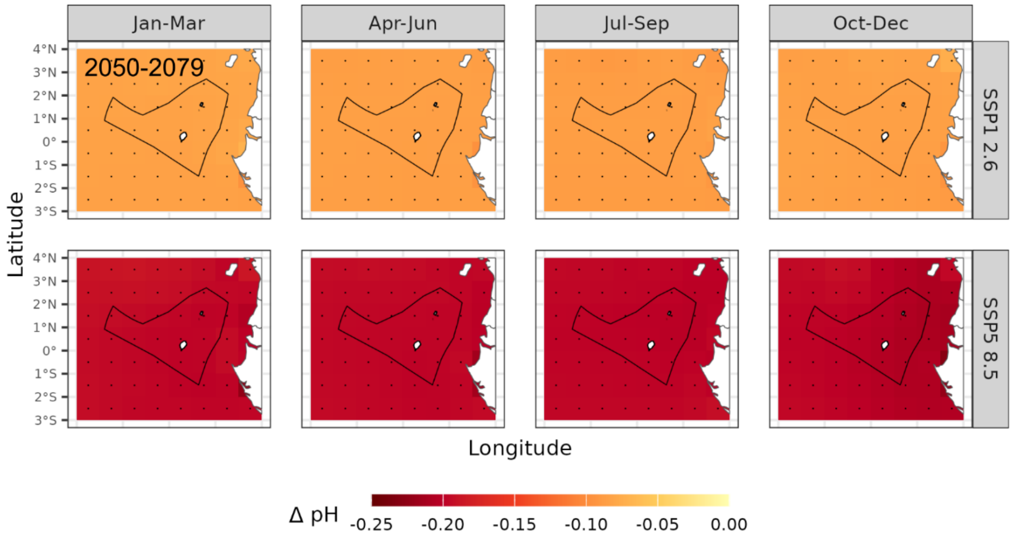


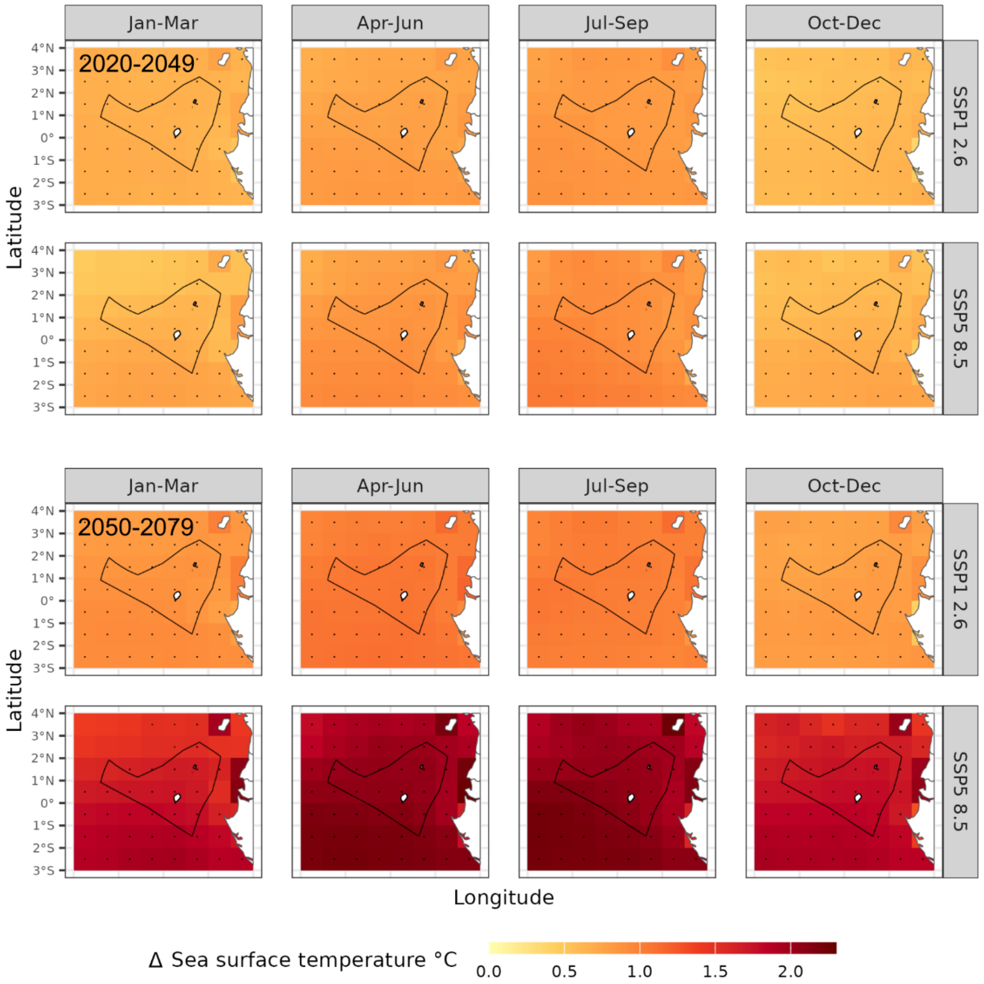




Fisheries – climate exposure

- Future climate trends:
  - *Projected temperature increases*
  - *Projected ocean acidification*
  - *Projected ocean deoxygenation*



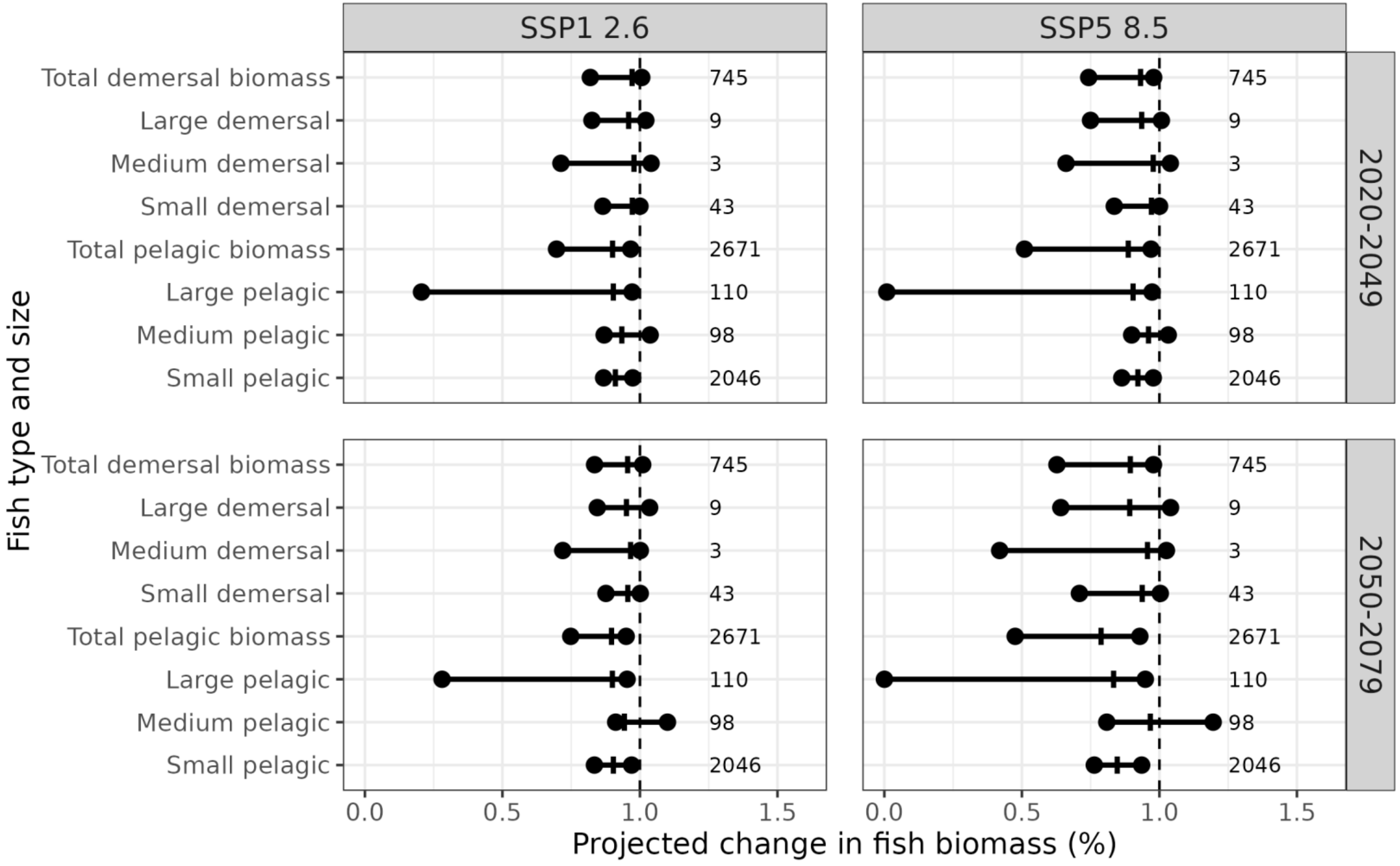


Fisheries – climate impact (mechanistic modelling)

ISIMIP

Fish biomass impact models:
- APECOSM
- BOATS
- EcoOcean
- DBPM
- FEISTY
- ZooMSS

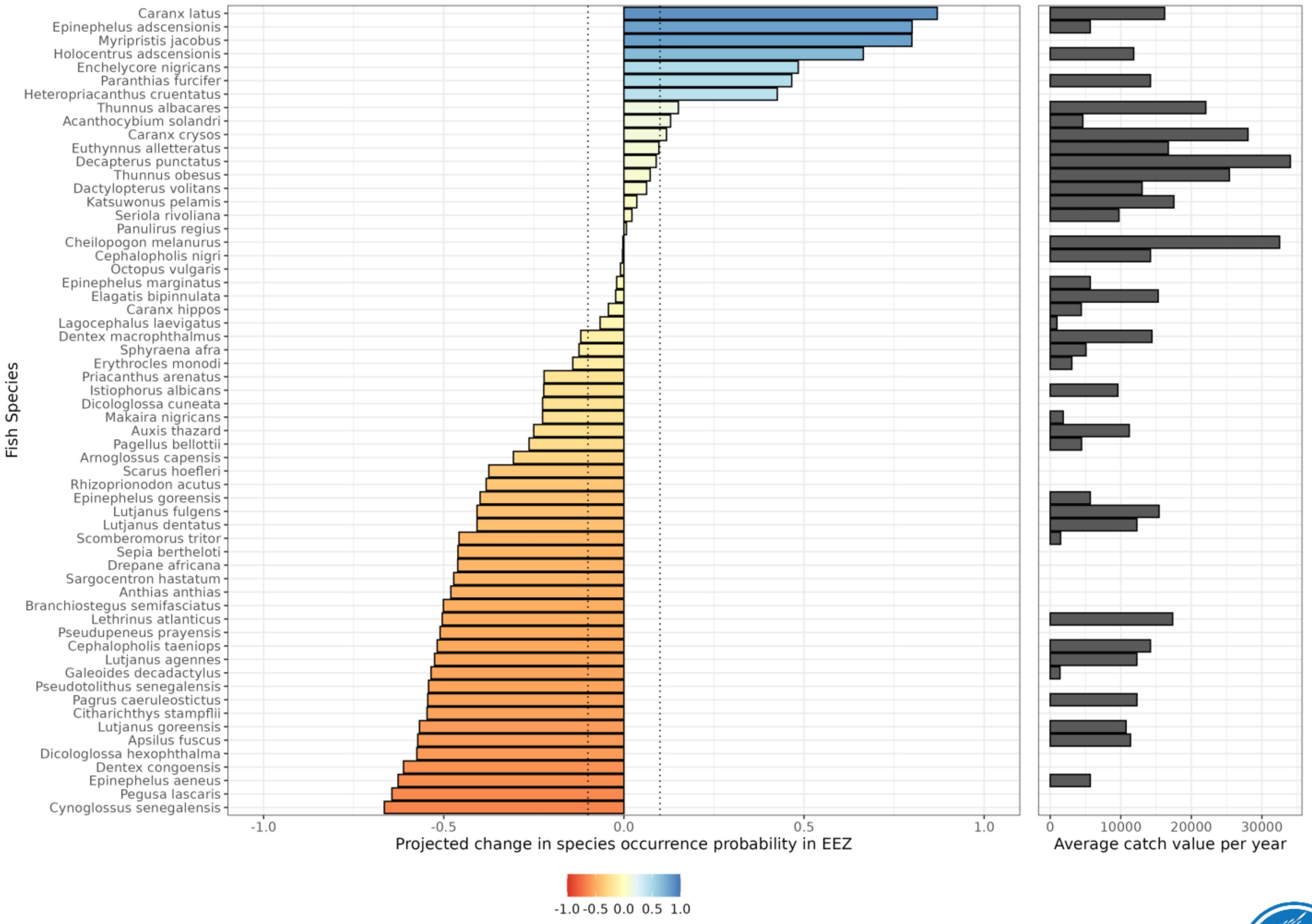
Climate models
- IPSL-CMIP6-LR
- GFDL-ESM4



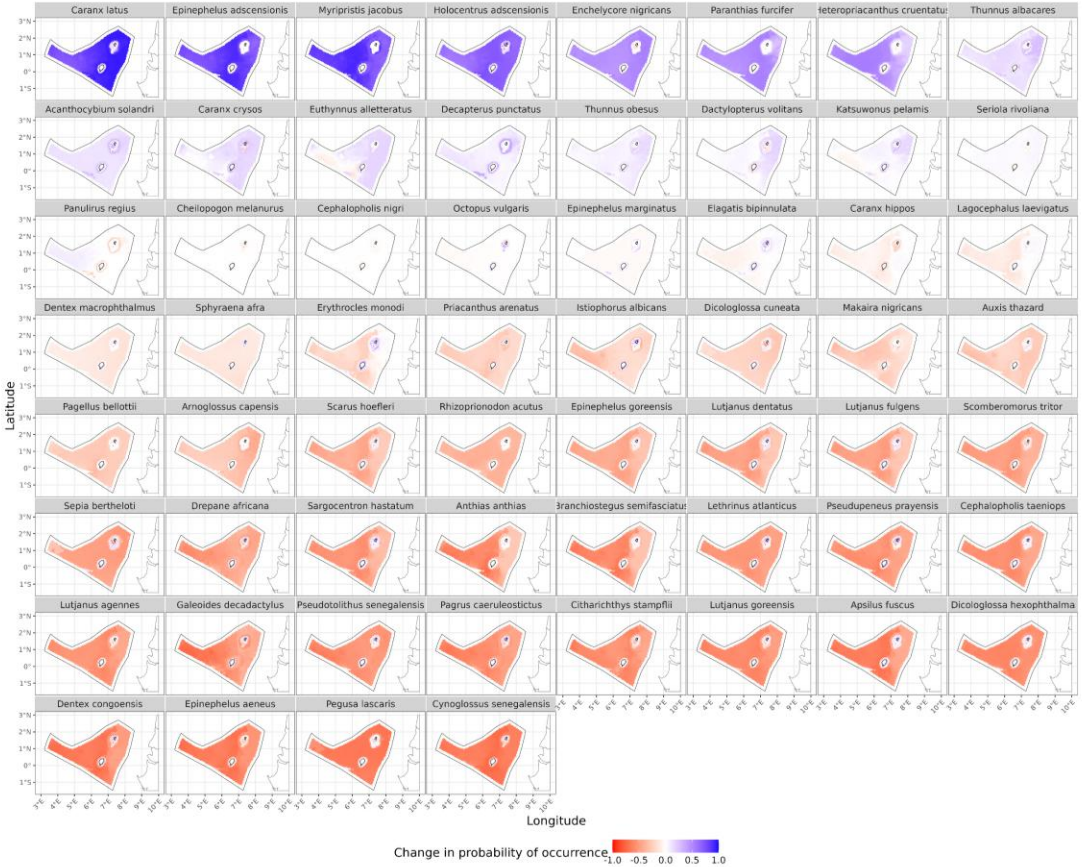
Fisheries – climate impact (empirical modelling)

- Fish occurrence probability is projected to vary considerably with climate change

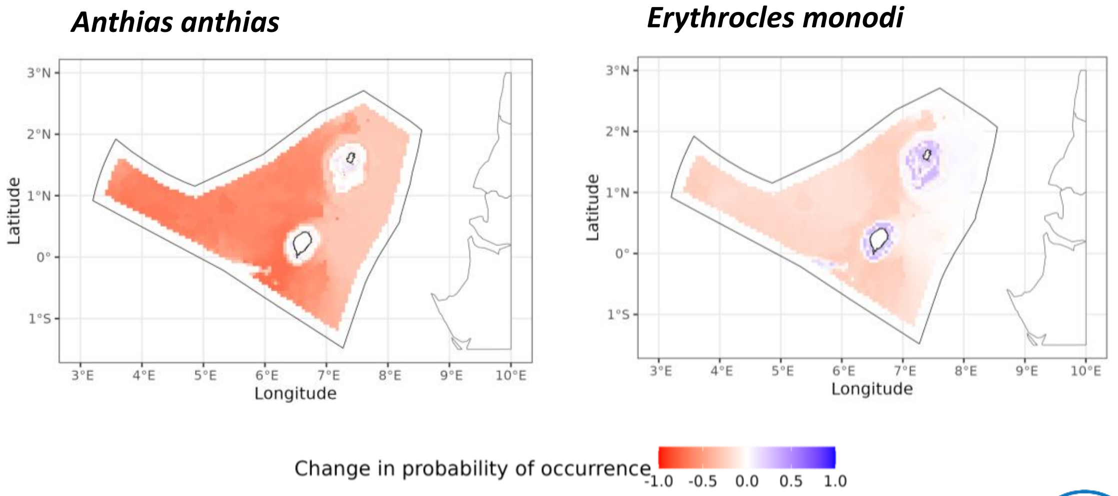
- With a 0.1 change cut-off, occurrence probability projected to:
  - Increase for **10** species
  - No change for **14** species
  - Decrease for **36** species



Fisheries – climate impact (empirical modelling)

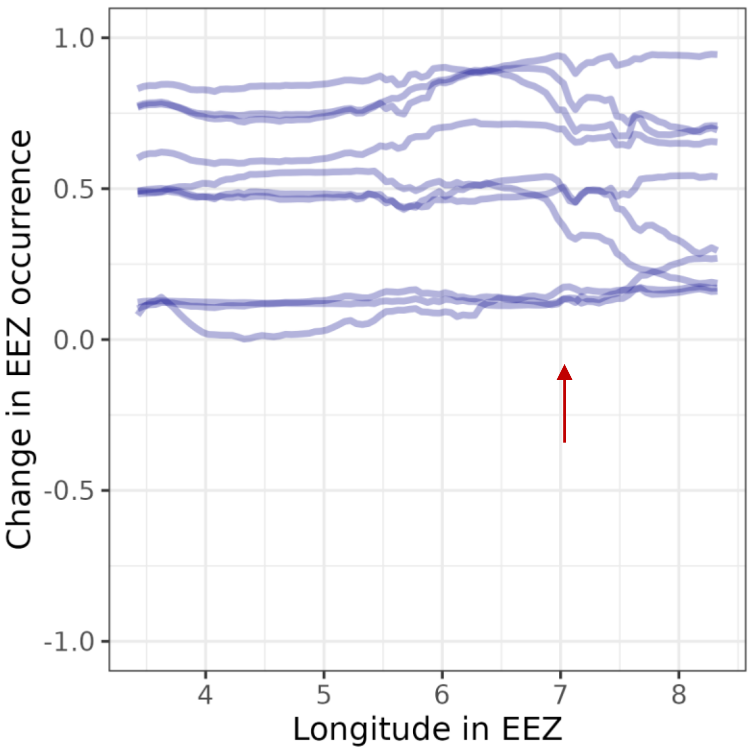


- Climate impact is greater in the west than east, consistent with mechanistic models
- Coastal waters are buffered from climate impact

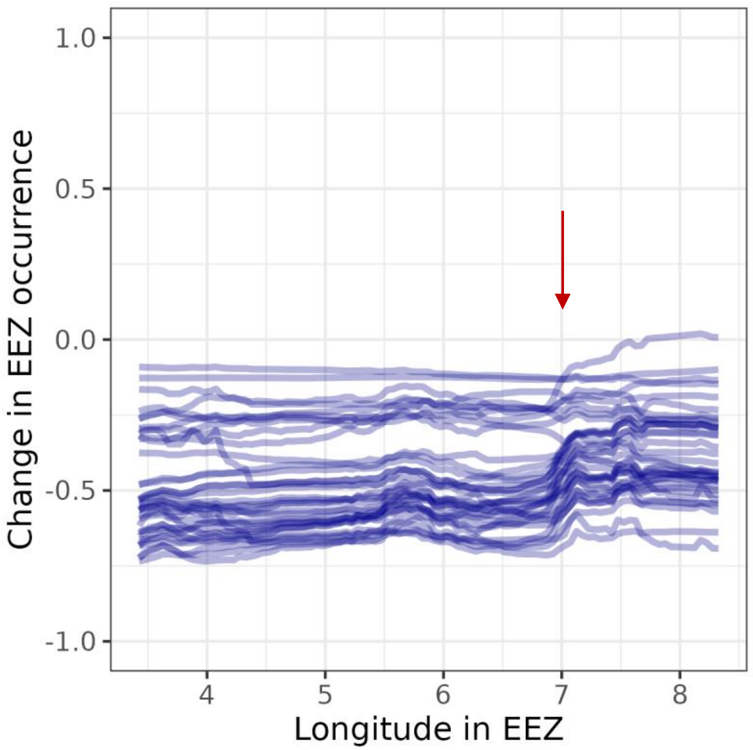


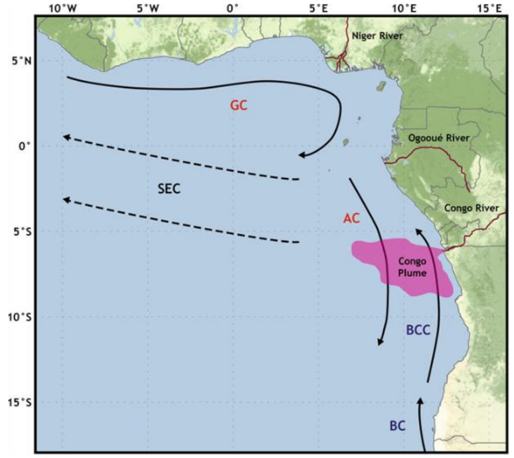
Fisheries – climate impact (empirical modelling)

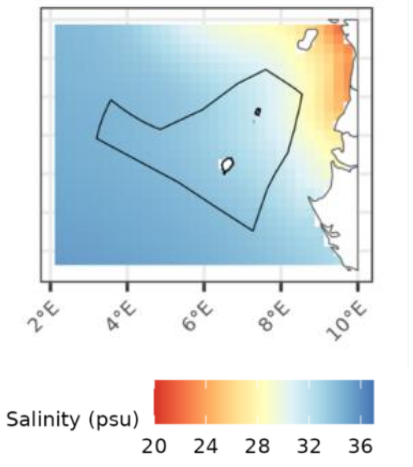
Positively affected by climate change



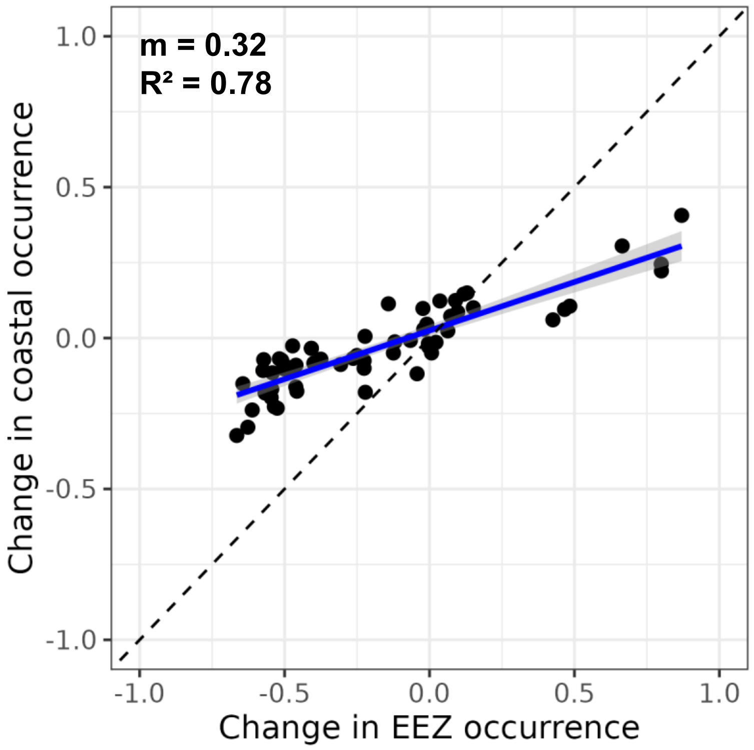
Negatively affected by climate change



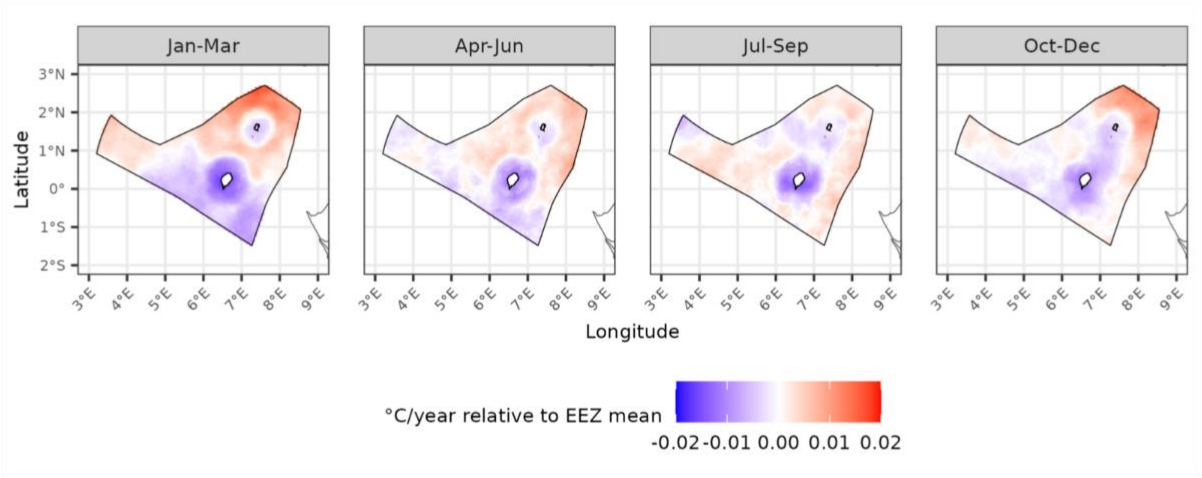




Fisheries – climate impact (empirical modelling)



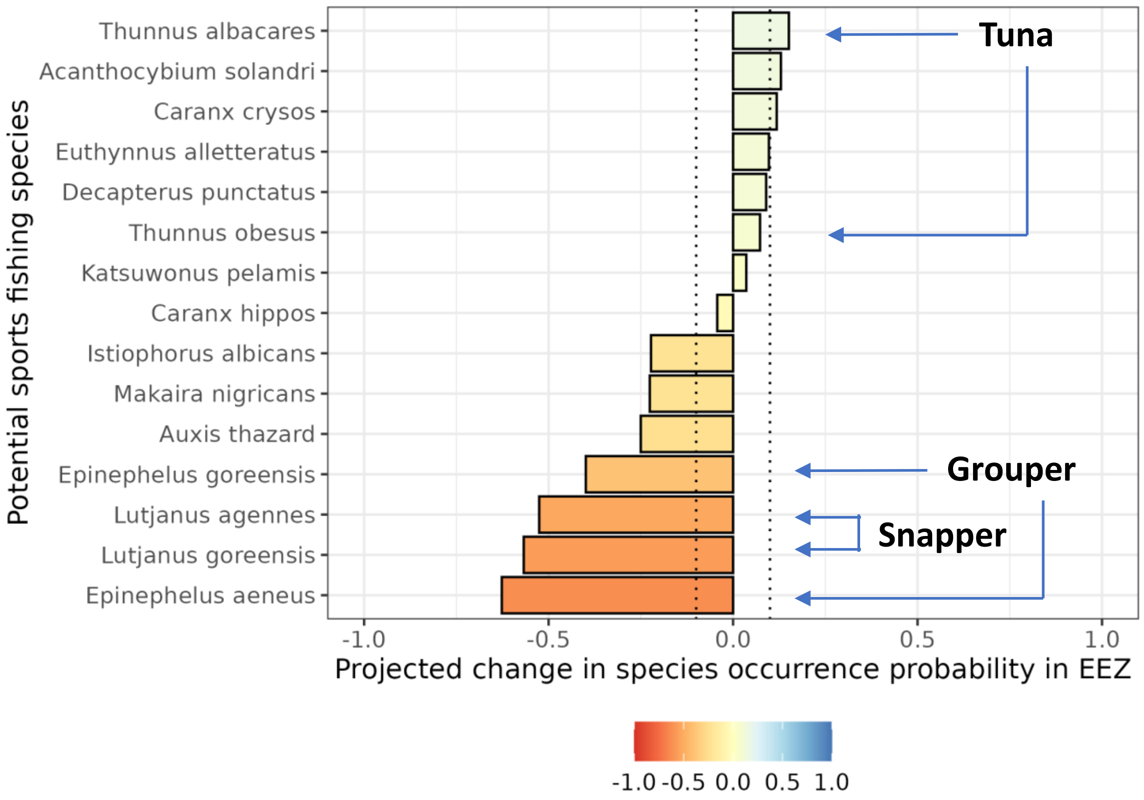
- Coastal buffering occurs across all species
- Result may be due to greater climatic stability near coast



Fisheries – climate impact (sports fishing)

- 15 fish species identified as candidates for sports fishing

- With a 0.1 change cut-off, presence probability projected to:
  - Increase for **3** species
  - No change for **5** species
  - Decrease for **7** species



Conclusions

- Future climate change is expected to affect both tourism and blue economy:
  - Sea level rise
  - Increased tourist discomfort
  - Reduction in EEZ fish biomass
  - Alterations to probability of pelagic and demersal occurrence in EEZ and species
- We recommend an adaptation strategy (outlined in a later session) to mitigate the impacts of climate change.

Merci
Thank you